DeepWAF-XAI: Master Source Data & Literature Benchmark Inventory

Underlying literature benchmarks, measured project evaluation, and comparability attributes.

System Name	Architecture / Method Type	Metric Name	Value (%)	Measurement Type	Source / Report Reference	Dataset or Evaluation Context	Notes on Direct Comparability & Scope
Traditional Rule-Based WAF (ModSecurity CRS)	Deterministic Regex Matching	Accuracy	82.40	Literature Benchmark	OWASP ModSecurity Core Rule Set Benchmark (CRS v3.3)	Real-world HTTP requests & OWASP test suite	Rule-based regex matching; high false-positive rate on complex inputs; lacks representation
Traditional Rule-Based WAF (ModSecurity CRS)	Deterministic Regex Matching	Precision	84.10	Literature Benchmark	OWASP ModSecurity Core Rule Set Benchmark (CRS v3.3)	Real-world HTTP requests & OWASP test suite	Susceptible to regex evasion and URL/unicode encoding tricks
Traditional Rule-Based WAF (ModSecurity CRS)	Deterministic Regex Matching	Recall	79.60	Literature Benchmark	OWASP ModSecurity Core Rule Set Benchmark (CRS v3.3)	Real-world HTTP requests & OWASP test suite	Frequent false negatives on novel zero-day attack variants
Traditional Rule-Based WAF (ModSecurity CRS)	Deterministic Regex Matching	F1-Score	81.80	Literature Benchmark	OWASP ModSecurity Core Rule Set Benchmark (CRS v3.3)	Real-world HTTP requests & OWASP test suite	Baseline harmonic mean across OWASP test partitions
Classical ML Baseline (SVM / RF)	N-Gram TF-IDF + Random Forest / SVM	Accuracy	88.70	Literature Benchmark	Standard Web Security Benchmark / N-gram Baseline	CSIC 2010 HTTP Dataset	Shallow statistical classification; lacks sequential contextual memory across variable-length
Classical ML Baseline (SVM / RF)	N-Gram TF-IDF + Random Forest / SVM	Precision	89.20	Literature Benchmark	Standard Web Security Benchmark / N-gram Baseline	CSIC 2010 HTTP Dataset	Evaluated on tabular n-gram token frequency matrices
Classical ML Baseline (SVM / RF)	N-Gram TF-IDF + Random Forest / SVM	Recall	86.50	Literature Benchmark	Standard Web Security Benchmark / N-gram Baseline	CSIC 2010 HTTP Dataset	Vulnerable to token-frequency poisoning and payload obfuscation
Classical ML Baseline (SVM / RF)	N-Gram TF-IDF + Random Forest / SVM	F1-Score	87.80	Literature Benchmark	Standard Web Security Benchmark / N-gram Baseline	CSIC 2010 HTTP Dataset	Standard machine learning baseline across web application firewall literature
BALNet (CNN Base Paper, 2026)	1D CNN + ADASYN Oversampling	Accuracy	91.30	Literature Benchmark	Springer Discover Applied Sciences (2026) 8:23	CSIC 2010 HTTP Dataset (ADASYN oversampled)	Evaluated on 3-class traffic; synthetically balanced with ADASYN; does not evaluate Path
BALNet (CNN Base Paper, 2026)	1D CNN + ADASYN Oversampling	Precision	92.00	Literature Benchmark	Springer Discover Applied Sciences (2026) 8:23	CSIC 2010 HTTP Dataset (ADASYN oversampled)	Reported benign precision 100.0%, SQLi 76.0%, and XSS 50.0%
BALNet (CNN Base Paper, 2026)	1D CNN + ADASYN Oversampling	Recall	90.80	Literature Benchmark	Springer Discover Applied Sciences (2026) 8:23	CSIC 2010 HTTP Dataset (ADASYN oversampled)	Reported benign recall 100.0%, SQLi 76.0%, and XSS 33.0%
BALNet (CNN Base Paper, 2026)	1D CNN + ADASYN Oversampling	F1-Score	91.40	Literature Benchmark	Springer Discover Applied Sciences (2026) 8:23	CSIC 2010 HTTP Dataset (ADASYN oversampled)	Weighted harmonic mean across ADASYN synthetic distribution
UniEmbed (Feature Fusion MLP, 2025)	Multi-Head Attention + Feature Fusion MLP	Accuracy	99.82	Literature Benchmark	Springer (2025), DOI: 10.1007/978-981-97-8899-6_11	Specialized payload dataset (binary SQLi / XSS)	Evaluated on isolated binary payload strings; does not evaluate Path Traversal or full multi-
UniEmbed (Feature Fusion MLP, 2025)	Multi-Head Attention + Feature Fusion MLP	Precision	99.80	Literature Benchmark	Springer (2025), DOI: 10.1007/978-981-97-8899-6_11	Specialized payload dataset (binary SQLi / XSS)	Binary classification benchmark on clean isolated tokens
UniEmbed (Feature Fusion MLP, 2025)	Multi-Head Attention + Feature Fusion MLP	Recall	99.86	Literature Benchmark	Springer (2025), DOI: 10.1007/978-981-97-8899-6_11	Specialized payload dataset (binary SQLi / XSS)	Near-perfect recall on targeted, isolated binary attack sets
UniEmbed (Feature Fusion MLP, 2025)	Multi-Head Attention + Feature Fusion MLP	F1-Score	99.83	Literature Benchmark	Springer (2025), DOI: 10.1007/978-981-97-8899-6_11	Specialized payload dataset (binary SQLi / XSS)	Payload-level binary classification score.
HawkPhish-DNN (HHO-DNN, 2025)	Harris Hawks Optimization + DNN	Accuracy	99.60	Literature Benchmark	Springer Cluster Computing (2025)	Phishing URL dataset (PhishTank, Alexa Top 1M)	Specialized solely for URL phishing; does not inspect HTTP payloads, SQL injection, or XSS
HawkPhish-DNN (HHO-DNN, 2025)	Harris Hawks Optimization + DNN	Precision	99.80	Literature Benchmark	Springer Cluster Computing (2025)	Phishing URL dataset (PhishTank, Alexa Top 1M)	High precision in separating legitimate domain names from phishing URLs
HawkPhish-DNN (HHO-DNN, 2025)	Harris Hawks Optimization + DNN	Recall	99.40	Literature Benchmark	Springer Cluster Computing (2025)	Phishing URL dataset (PhishTank, Alexa Top 1M)	High phishing intercept rate under HHO hyperparameter search
HawkPhish-DNN (HHO-DNN, 2025)	Harris Hawks Optimization + DNN	F1-Score	99.60	Literature Benchmark	Springer Cluster Computing (2025)	Phishing URL dataset (PhishTank, Alexa Top 1M)	Domain-specific phishing benchmark score.
Proposed DeepWAF-XAI	Stacked BiLSTM + Attention + XAI	Accuracy (Overall)	92.59	Project-Measured Result	Project Evaluation Output (results.json)	Stratified held-out test split (n = 19,776 / 197,756)	Measured on unaugmented natural imbalanced 5-class web traffic (18,311 / 19,776
Proposed DeepWAF-XAI	Stacked BiLSTM + Attention + XAI	Precision (Weighted)	97.14	Project-Measured Result	Project Evaluation Output (results.json)	Stratified held-out test split (n = 19,776 / 197,756)	Weighted by class support; Macro Precision is 58.23% due to natural minority imbalance
Proposed DeepWAF-XAI	Stacked BiLSTM + Attention + XAI	Recall (Weighted)	92.59	Project-Measured Result	Project Evaluation Output (results.json)	Stratified held-out test split (n = 19,776 / 197,756)	Weighted by class support; Macro Recall is 89.43%
Proposed DeepWAF-XAI	Stacked BiLSTM + Attention + XAI	F1-Score (Weighted)	94.32	Project-Measured Result	Project Evaluation Output (results.json)	Stratified held-out test split (n = 19,776 / 197,756)	Weighted by class support; Macro F1-Score is 65.93%
Proposed DeepWAF-XAI	Normal Traffic (LEGAL)	Accuracy	92.40	Project-Measured Result	classification_report_test.txt (support: 18,290)	Stratified held-out test split	16,900 / 18,290 correctly classified normal requests
Proposed DeepWAF-XAI	Normal Traffic (LEGAL)	Precision	99.92	Project-Measured Result	classification_report_test.txt (support: 18,290)	Stratified held-out test split	Near-zero false alarms on legitimate business traffic
Proposed DeepWAF-XAI	Normal Traffic (LEGAL)	Recall	92.40	Project-Measured Result	classification_report_test.txt (support: 18,290)	Stratified held-out test split	Legitimate request passage rate.
Proposed DeepWAF-XAI	Normal Traffic (LEGAL)	F1-Score	96.01	Project-Measured Result	classification_report_test.txt (support: 18,290)	Stratified held-out test split	High reliability on primary traffic volume.
Proposed DeepWAF-XAI	SQL Injection (SQL)	Accuracy	77.58	Project-Measured Result	classification_report_test.txt (support: 165)	Stratified held-out test split	128 / 165 correctly intercepted SQL injection attempts
Proposed DeepWAF-XAI	SQL Injection (SQL)	Precision	14.55	Project-Measured Result	classification_report_test.txt (support: 165)	Stratified held-out test split	Lower precision due to severe natural class imbalance (165 SQL vs 18,290 Normal)
Proposed DeepWAF-XAI	SQL Injection (SQL)	Recall	77.58	Project-Measured Result	classification_report_test.txt (support: 165)	Stratified held-out test split	Solid recall boundary; mitigated further by secondary DB firewall
Proposed DeepWAF-XAI	SQL Injection (SQL)	F1-Score	24.50	Project-Measured Result	classification_report_test.txt (support: 165)	Stratified held-out test split	Harmonic score under natural production imbalance
Proposed DeepWAF-XAI	Cross-Site Scripting (XSS)	Accuracy	95.94	Project-Measured Result	classification_report_test.txt (support: 789)	Stratified held-out test split	757 / 789 correctly intercepted XSS attack scripts
Proposed DeepWAF-XAI	Cross-Site Scripting (XSS)	Precision	56.24	Project-Measured Result	classification_report_test.txt (support: 789)	Stratified held-out test split	Exceeds published BALNet XSS precision (50.0%)
Proposed DeepWAF-XAI	Cross-Site Scripting (XSS)	Recall	95.94	Project-Measured Result	classification_report_test.txt (support: 789)	Stratified held-out test split	Outperforms BALNet published XSS recall (33.0%) by +62.94%
Proposed DeepWAF-XAI	Cross-Site Scripting (XSS)	F1-Score	70.91	Project-Measured Result	classification_report_test.txt (support: 789)	Stratified held-out test split	Strong balance on script execution payload detection

Proposed DeepWAF-XAI	Path Traversal (SHELL)	Accuracy	81.25	Project-Measured Result	classification_report_test.txt (support: 32)	Stratified held-out test split	26 / 32 correctly intercepted path traversal requests
Proposed DeepWAF-XAI	Path Traversal (SHELL)	Precision	28.89	Project-Measured Result	classification_report_test.txt (support: 32)	Stratified held-out test split	Extreme minority class (32 samples out of 19,776)
Proposed DeepWAF-XAI	Path Traversal (SHELL)	Recall	81.25	Project-Measured Result	classification_report_test.txt (support: 32)	Stratified held-out test split	Effective containment of directory traversal and LFI payloads
Proposed DeepWAF-XAI	Path Traversal (SHELL)	F1-Score	42.62	Project-Measured Result	classification_report_test.txt (support: 32)	Stratified held-out test split	Literature DL models (BALNet, UniEmbed) report N/A on Path Traversal
Proposed DeepWAF-XAI	Phishing URLs (PHISHING)	Accuracy	100.00	Project-Measured Result	classification_report_test.txt (support: 500)	Stratified held-out test split	500 / 500 correctly intercepted phishing URLs.
Proposed DeepWAF-XAI	Phishing URLs (PHISHING)	Precision	91.58	Project-Measured Result	classification_report_test.txt (support: 500)	Stratified held-out test split	High precision distinguishing phishing URLs from legal queries
Proposed DeepWAF-XAI	Phishing URLs (PHISHING)	Recall	100.00	Project-Measured Result	classification_report_test.txt (support: 500)	Stratified held-out test split	Perfect detection recall on held-out phishing test partition
Proposed DeepWAF-XAI	Phishing URLs (PHISHING)	F1-Score	95.60	Project-Measured Result	classification_report_test.txt (support: 500)	Stratified held-out test split	Competitive with specialized HawkPhish-DNN architecture